

DEMAND ESTIMATION FOR PUBLIC TRANSPORTATION IN COLOMBIA

Daniel Toro González*

Extended Abstract

The design of the traditional public transportation system in most of the middle size cities of Colombia generates major problems in mobility. The structure in which the remuneration of the driver is related to the number passengers he transports has promoted low vehicle speeds, long travel times, irregular service schedules, excess supply of vehicles and, in general, low quality service. To deal with these problems, people in many urban areas of intermediate cities has started to use informal or unlicensed transportation modes such as motorcycle taxi, and collective transportation vehicles, both, at a competitive market tariffs. In fact, the fleet of buses has decline in recent years because the shift in demand preferences towards unlicensed transportation modes (Escobar García 2008). The determinants of this demand shift have been studied in some cities such as Cartagena, and are related mainly with travel time, availability and tariff (Toro-González, Alvis and Arellano 2005).

Public transportation systems have not been studied sufficiently in Colombia. Some efforts have been made to analyze demand in specific cities, or some other, using a macroeconomic perspective of the public transportation role and impact on the economy. Hence, due to the lack of knowledge about the consumption patterns for the service, an analysis of the public transportation market is necessary to establish the demand parameters that govern the general behavior of the consumers, help to predict the market, and provide instruments to design of better public policies.

* Associate Professor at Universidad Tecnológica de Bolívar, Cartagena de Indias, Colombia. Abstract submitted to hEART 2013 2nd Symposium of the European Association for Research in Transportation, Sweden.

This study presents an estimation of the demand for licensed public transportation in Colombia during 2000-2010 analyzing the main determinants to explain the market shares of licensed public transportation modes.

The article utilizes a city level aggregated market share model (S. T. Berry 1994), using each of the public transportation modes as a differentiated service. The model uses information for 22 cities and metropolitan areas collected by the Colombian National Statistics Bureau, DANE. The modeling strategy allowed us to identify the main determinants in the demand for licensed public transportation in Colombia. It also estimates the overall price, income and speed elasticities, and offers a tool to build counterfactuals scenarios to predict the market share impacts of changes in tariff or speed by any transportation mode.

Our results suggest that the demand for transportation in Colombia is price inelastic and responds positively in speed increase. We also find a negative income elasticity which indicates that licensed public transportation is an inferior good. Considering these above results and increased income concentration in Colombia last decade, we recommend the implementation of subsidized Bus Rapid Transit (BRT) systems to improve the quality of public transportation systems and income equality simultaneously.

Bibliography

Berry, Steven T. "Estimating discrete-choice models of product differentiation." *RAND Journal of Economics* 25, no. 2 (Summer 1994): 242-262.

Escobar García, Diego Alexander. "Instrumentos y Metodología de Planes de Movilidad y Transporte en las Ciudades Medias Colombianas." *Tesis Doctoral*. Electronic Document. Universidad Politécnica de Cataluña. Universidad Politécnica de Cataluña, 2008.

Toro-González, Daniel, Jorge Alvis, and William Arellano. "Transporte público en Cartagena: ¿Qué factores determinan las preferencias de los usuarios?" *Economía & Región* (Universidad Tecnológica de Bolívar) 2, no. 3 (July 2005): 7-54.